1 · EOQ and EPQ

Variables: D demand rate · S cost per order/setup · C unit cost · p selling price · i holding rate (%/yr) · $H = iC$ holding cost per unit per year · Q lot size · P production rate ($P > D$) · L lead time · T cycle · n orders (runs) per year · T_p uptime · T_d downtime · TVC total variable cost/yr · TC total cost/yr · π profit/yr.

EOQ assumptions: demand & lead time known and constant; instantaneous receipt; constant price; holding on average inventory; constant order cost; no backorders. Trade-off: ordering (economies of scale) vs. holding. **EPQ:** production at P while demand consumes at $D \to$ stock rises at $P - D$, never reaches Q. EPQ > EOQ; $P \to \infty$ gives EOQ.

	EOQ	EPQ
TVC(Q)	$S\frac{D}{Q} + H\frac{Q}{2}$	$S\frac{D}{Q} + H\frac{Q}{2}\left(1 - \frac{D}{P}\right)$
TC(Q)	$CD + \text{TVC}(Q)$	
$\pi(Q)$	$(p-C)D - S\frac{D}{Q} - H\frac{Q}{2}$	$(p-C)D - S\frac{D}{Q} - H\frac{Q}{2}\left(1 - \frac{D}{P}\right)$
π^*	$(p-C)D - \sqrt{2DSH}$	$(p-C)D - \sqrt{2DSH(1-D/P)}$
FOC	$-S\frac{D}{Q^2} + \frac{H}{2} = 0$	$-S\frac{D}{Q^2} + \frac{H}{2}\left(1 - \frac{D}{P}\right) = 0$
Q^*	$\sqrt{\frac{2DS}{H}}$	$\sqrt{\frac{2DS}{H(1-D/P)}}$
TVC(Q^*)	$\sqrt{2DSH}$	$\sqrt{2DSH(1-D/P)}$
$I_{\max}$	Q	$T_p(P-D) = Q(1-D/P)$
$\bar{I}$ avg	$Q/2$	$I_{\max}/2$ (not $Q/2$)
T_p	0	Q/P
T_d	Q/D	$I_{\max}/D$
T	Q/D	$T_p + T_d = Q/D$
n	$D/Q = 1/T$	
T^*	$\sqrt{\frac{2S}{DH}}$	$\sqrt{\frac{2S}{DH(1-D/P)}}$
n^*	$\sqrt{\frac{DH}{2S}}$	$\sqrt{\frac{DH(1-D/P)}{2S}}$
ρ util.	0	$D/P = T_p/T$ (need $D < P$)
$1-\rho$ idle	–	$1 - D/P = T_d/T$; idle time/yr $= (1-D/P) \times 1$ yr

- **Profit:** with p and C fixed, $(p-C)D$ is a constant $\to \max \pi \Leftrightarrow \min$ TVC, same Q^*. Price breaks or $p(Q)$ change this (use TC/π per range).
- At Q^*: ordering cost = holding cost = TVC*/2. EPQ downtime/yr $= nT_d = (1-\rho)$ yr. Savings $= 1 - \text{TVC}(Q^*)/\text{TVC}(Q_{\text{current}})$.
- **Reorder point** (on-hand trigger, D, L same unit). $L \le T$: ROP $= DL$. $L > T$: inventory position still triggers at DL, but $\lfloor L/T \rfloor$ orders are in transit:

$$\text{ROP}_{\text{on-hand}} = DL - \lfloor L/T \rfloor Q$$

- EPQ: DL is the on-hand trigger only if $L \le T_d$; if $T_d < L \le T$ the reorder falls during the run and on-hand $= (P-D)(T-L)$.

2 · Sensitivity and robustness

$$r = \frac{\text{TVC}(Q)}{\text{TVC}(Q^*)} = \frac{1}{2}\left(\frac{Q^*}{Q} + \frac{Q}{Q^*}\right) \qquad \%\text{ increase} = (r-1) \times 100$$

- Steps: compute $Q^* \to$ take the other Q (given / rounded / policy) $\to r \to (r-1)100\%$.
- Benchmarks: $Q/Q^* = 4/3$ or $3/4 \to$ +4.2% · $\sqrt{2} \to$ +6.1% · 2 or ½ → +25%. TVC is flat near Q^*.
- **Scaling:** $Q^* \propto \sqrt{DS/H}$, TVC* $\propto \sqrt{DSH}$, $n^* \propto \sqrt{DH/S}$. A parameter $\times k \to$ multiply by $\sqrt{k}$ if it is in the numerator, divide by $\sqrt{k}$ if in the denominator. E.g. $S \times 4$: Q^* and TVC* ×2. To halve the lot, S must fall to $S/4$.
- **Mis-estimated input** (D or S off by factor k, or H off by $1/k$): $Q = \sqrt{k}Q^*$, penalty $r = \frac{1}{2}\left(\sqrt{k} + 1/\sqrt{k}\right)$ (e.g. $k = 2 \to$ +6.1%).

3 · Power-of-two policies

Variables: T^* optimal interval · T_B base period (1 week = 1/52 yr) · $T = 2^k T_B$, $k = 0, 1, 2, \ldots$ · $\text{TVC}(T) = S/T + HDT/2$ (with $Q = DT$).

1. $T^* = Q^*/D$ (convert to base periods).
2. **Range test:** choose the k with $2^k T_B \in \left[T^*/\sqrt{2},\ \sqrt{2}T^*\right]$ (exactly one exists).
   Equivalent: smallest k with $\text{TVC}\left(2^k T_B\right) \le \text{TVC}\left(2^{k+1}T_B\right)$.

k	2^k	In $\left[T/\sqrt{2}, T\sqrt{2}\right]$
0	0	no
1	2	no
2	4	**yes**
3	8	no

3. Cost ratio $r = \frac{1}{2}\left(\frac{T^*}{T} + \frac{T}{T^*}\right) \le \frac{1}{2}\left(\sqrt{2} + \frac{1}{\sqrt{2}}\right) = 1.0607 \to$ at most +6.1%.
4. Lot $Q = DT$; % increase $= (r-1) \times 100$.

4 · All-unit quantity discounts

Variables: Break points $q_0 = 0 < q_1 < \ldots < q_m = \infty$ · if $q_i \le Q < q_{i+1}$ **every** unit costs C_i · $C_0 \ge C_1 \ge \ldots$ · $H_i = hC_i$ changes with price.

$$\text{TC}_i(Q) = \frac{D}{Q}S + \frac{Q}{2}hC_i + DC_i \quad q_i \le Q < q_{i+1} \qquad Q_i = \sqrt{\frac{2DS}{hC_i}}$$

1. Start at the **lowest** price. If Q_i is feasible ($q_i \le Q_i < q_{i+1}$): candidate Q_i, **stop** (higher prices cannot beat it).
2. If $Q_i < q_i$: candidate = q_i (the break point); move to the next higher price and repeat.
3. Q^* = candidate with the lowest TC_i. Cycle inventory $= Q^*/2$.

Must include purchase cost DC_i (it differs across ranges): compare TC, not TVC. Read break points carefully: "1,001–2,000" → $q_i = 1\,001$.

5 · Aggregation (joint ordering)

Variables: m products · S common cost per order (truck/transport) · s_i product-specific cost (paid when i is in the order) · D_i, C_i, h, $H_i = hC_i$ · n orders/yr · K truck capacity (units per order).

No aggregation (each product alone, $S_i' = S + s_i$)

$$Q_i^* = \sqrt{\frac{2D_iS_i'}{H_i}} \quad n_i^* = \sqrt{\frac{D_iH_i}{2S_i'}} \quad \text{TVC}_{\text{no}} = \sum_i \sqrt{2D_iS_i'H_i}$$

Complete aggregation (all products every order, $S' = S + \sum_i s_i$)

$$n = D_i/Q_i \text{ (same for all)} \quad \text{TVC}(n) = nS' + \frac{\sum_i H_iD_i}{2n}$$

$$n^* = \sqrt{\frac{\sum_i D_iH_i}{2S'}} \quad Q_i^* = \frac{D_i}{n^*} \quad \text{TVC}_{\text{agg}} = \sqrt{2S'\sum_i D_iH_i} = 2n^*S'$$

- Savings $= \left(1 - \text{TVC}_{\text{agg}}/\text{TVC}_{\text{no}}\right) \times 100\%$.
- Report per product: Q_i, n, cycle inventory $Q_i/2$, days of inventory $= Q_i/(2D_i) \times 365$.

Infeasible solution: capacity K binds

TVC is convex in Q (or n) → if the unconstrained optimum breaks capacity, the best feasible point is **on the boundary** (full truck).

Case	If infeasible → fix and recompute
No aggr. ($Q_i^* > K$)	$Q_i = K,\ n_i = D_i/K$ $\text{TVC}_i = (D_i/K)(S + s_i) + H_iK/2$ (not $\sqrt{2D_iS_i'H_i}$)
Aggregated ($\sum_i D_i/n^* > K$)	$n = \sum_i D_i/K > n^*,\ Q_i = D_i/n = KD_i/\sum_j D_j$ $\text{TVC} = nS' + \sum_i H_iD_i/(2n)$ (not $2nS'$)

- Savings then use the **constrained** costs on both sides. Order-cost share ≠ holding share → check by computing both terms.

6 · Newsvendor (single period)

Variables: D demand, $N(\mu, \sigma)$ · Q order qty · p price · c unit cost · s salvage per leftover · g goodwill per unit short (0 if not given) · C_u underage (1 too few) · C_o overage (1 too many) · α critical ratio · F CDF · Φ, φ std. normal CDF, pdf · $L(z)$ std. normal loss · θ fraction by which σ is cut.

Order Q once, before demand → pay cQ → demand → sell $\min(Q, D)$ at p → leftovers at s → each shortage costs g. If unsure of C_u, C_o: profit lost by having 1 fewer / 1 more unit.

$$C_u = p - c + g \qquad C_o = c - s$$

$$\text{Marginal unit: } C_oF(Q) = C_u(1 - F(Q))$$

$$F(Q^*) = P(D \le Q^*) = \alpha = \frac{C_u}{C_u + C_o} = \frac{p - c + g}{p - s + g}$$

$$z^* = \Phi^{-1}(\alpha), \quad Q^* = \mu + z^*\sigma$$

$$\text{round } Q \text{ up} \cdot \alpha < 0.5 \Rightarrow z^* < 0 \Rightarrow Q^* < \mu$$

Performance for any Q (first: $z = (Q-\mu)/\sigma$)

	$z = (Q-\mu)/\sigma$ (use the **rounded** Q) → look up $\Phi(z)$, $L(z)$, $\varphi(z)$
In-stock (Type I SL)	$P(D \le Q) = \Phi(z)$ (= α at Q^*) · counts **seasons**
Stockout prob.	$P(D > Q) = 1 - \Phi(z)$
E[short] (lost sales)	$E(D-Q)^+ = \sigma L(z), \quad L(z) = \varphi(z) - z(1 - \Phi(z))$
E[sales]	$E[\min(D,Q)] = \mu - E[\text{short}]$ (always $< \mu$)
E[left]	$E(Q-D)^+ = Q - E[\text{sales}] = Q - \mu + E[\text{short}] = \sigma(z + L(z))$
Identities	$E[\text{sales}] + E[\text{short}] = \mu \quad \cdot \quad E[\text{sales}] + E[\text{left}] = Q$
Fill rate (Type II)	$E[\text{sales}]/\mu = 1 - E[\text{short}]/\mu$ · counts **units** → usually > in-stock
Target fill rate FR	$L(z) = (1 - \text{FR})\mu/\sigma \to z$ from loss table (take larger z) $\to Q = \mu + z\sigma$
Expected profit (any Q)	$\pi = pE[\text{sales}] + sE[\text{left}] - gE[\text{short}] - cQ$ $= (p-c)\mu - C_uE[\text{short}] - C_oE[\text{left}]$
Mismatch cost	$C(Q) = C_uE[\text{short}] + C_oE[\text{left}] \quad \to \quad \pi = (p-c)\mu - C(Q)$
Max profit (only at Q^*)	$\pi(Q^*) = (p-c)\mu - (C_u + C_o)\varphi(z^*)\sigma$ $\varphi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$
Cut σ by θ	$\theta = \frac{\sigma_{\text{old}} - \sigma_{\text{new}}}{\sigma_{\text{old}}} = 1 - \frac{\sigma_{\text{new}}}{\sigma_{\text{old}}}$ $\Delta\pi = \theta(C_u + C_o)\varphi(z^*)\sigma_{\text{old}} = (C_u + C_o)\varphi(z^*)(\sigma_{\text{old}} - \sigma_{\text{new}})$ (z^* unchanged; profit linear in σ; φ = pdf, not Φ)

Mismatch cost

- $(p-c)\mu$: crystal-ball profit − knowing D, you order exactly D and earn $p - c$ on every unit.
- **Too few** (lost sale): each costs $C_u = p - c + g$, expect $E[\text{short}] \to C_uE[\text{short}]$.
- **Too many** (leftover): each costs $C_o = c - s$, expect $E[\text{left}] \to C_oE[\text{left}]$.
- $C(Q) = C_uE[\text{short}] + C_oE[\text{left}]$ = expected cost of guessing wrong → $\pi = (p-c)\mu - C(Q)$: maximizing profit ⇔ minimizing mismatch cost.
- **At Q^*** (proof): $E[\text{left}] = \sigma(z^* + L(z^*))$ and $1 - \Phi(z^*) = C_o/(C_u + C_o) \to C(Q^*) = \sigma[(C_u + C_o)L(z^*) + C_oz^*] = (C_u + C_o)\varphi(z^*)\sigma$ = constant × σ.

Value of cutting σ

- α and z^* do not depend on $\sigma \to C(Q^*)$ and π^* are **linear** in σ; with $\sigma = 0$ you earn the crystal-ball profit.
- $\theta = 1 - \sigma_{\text{new}}/\sigma_{\text{old}} \to \Delta\pi = \theta(C_u + C_o)\varphi(z^*)\sigma_{\text{old}} = (C_u + C_o)\varphi(z^*)(\sigma_{\text{old}} - \sigma_{\text{new}})$ = max to pay for a forecast tool / quick response.
- Use the C_u, C_o of **the setting** (e.g. 2nd supplier: $C_u = c_P - c_C$) → different z^*, $\varphi(z^*)$, so a fast supplier makes better forecasts worth less. φ = pdf, not Φ.

Variants

- **Discrete demand table:** Q^* = smallest Q with $F(Q) \ge \alpha$ (cumulate probabilities). $E[\text{short}] = \sum_{d>Q}(d-Q)P(d)$; then the same chain.
- **Goodwill:** $C_u = p - c + g$ and $-gE[\text{short}]$ in π. Compare scenarios with the same g.
- **Fixed order cost** K: Q^* unchanged; profit $\pi(Q^*) - K$; order only if $> \pi(0) = -g\mu$ (with $g = 0$: if > 0).
- **Sensitivity:** $s \uparrow \Rightarrow C_o \downarrow \Rightarrow Q^* \uparrow \cdot g \uparrow$ or $p \uparrow \Rightarrow Q^* \uparrow \cdot c \uparrow \Rightarrow Q^* \downarrow \cdot \sigma \uparrow$: $Q^* \uparrow$ if $z^* > 0$, $\downarrow$ if $z^* < 0$; $\pi^* \downarrow$ always.

Uniform demand $D \sim U[a, b]$ (exact, no tables)

$$\mu = \frac{a+b}{2} \quad \sigma = \frac{b-a}{\sqrt{12}} \quad F(Q) = \frac{Q-a}{b-a}$$

$$Q^* = a + \alpha(b-a) \qquad P(\text{short}) = 1 - \alpha = \frac{b-Q}{b-a}$$

E[short]	$(b-Q)^2/(2(b-a))$
E[left]	$(Q-a)^2/(2(b-a))$
E[sales]	$\mu - E[\text{short}] = Q - E[\text{left}]$
Fill rate	$1 - E[\text{short}]/\mu$

Profit	same chain: $\pi = (p-c)\mu - C_u E[\text{short}] - C_o E[\text{left}]$

- Valid for $a \le Q \le b$. If $Q \le a$: nothing left over, $E[\text{short}] = \mu - Q$. If $Q \ge b$: nothing short.
- **Second run on residual demand**
  - $U[a,b]$: $Q_2^* = a + \alpha(b-a)$
  - max short $= (1-\alpha)(b-a)$
  - max left $= \alpha(b-a)$
  - $P(R > Q_2^*) = \frac{b-Q_2^*}{b-a} = 1-\alpha$

Backup supplier (China c_C before season · Portugal c_P after D known)

A unit short can be lost (costs $p+g$) or bought from Portugal (costs c_P) → take the cheaper. Portugal has no demand risk; China is cheap but must be ordered blind.

$$C_u = \min(p+g,\, c_P) - c_C \qquad C_o = c_C - s$$

Condition	Decision
$c_P \ge p+g$	**All from China** · Portugal never used (dearer than losing the sale) → plain newsvendor, $C_u = p - c_C + g$
$c_C < c_P < p+g$	**Both** · China Q_C from α below, Portugal tops up the shortfall
$c_P \le c_C$	**All from Portugal** · $Q_C = 0$ ($C_u \le 0$); $\pi = (p-c_P)\mu$, no risk

Both suppliers ($c_C < c_P < p+g$) — recalculate the chain:

$$C_u = c_P - c_C \quad \alpha = \frac{c_P - c_C}{c_P - s} \quad z = \Phi^{-1}(\alpha) \quad Q_C = \mu + z\sigma$$
$$E[\text{short}] = \sigma L(z) = E[\text{Portugal order}] \quad E[\text{left}] = Q_C - \mu + E[\text{short}]$$
$$\pi = p\mu - c_C Q_C - c_P E[\text{short}] + sE[\text{left}]$$

- $E[\text{sales}] = \mu$ (every unit is served) → fill rate and in-season service = 100%; g **drops out** of π (no lost sales).
- Max one-time fee for Portugal $= \pi_{\text{both}} - \pi_{\text{China only}}$. Goodwill $g \uparrow \to \pi_{\text{China only}} \downarrow$, π_{both} unchanged → Portugal worth more.
- Fixed cost K per China order: Q_C unchanged; order from China only if $\pi_{\text{both}}(Q_C) - K > (p-c_P)\mu$ (Portugal-only profit).
- Value of cutting σ by θ: $\theta(c_P - s)\varphi(z)\sigma$ (smaller than China-only, since Portugal already absorbs risk).

7 · Supply-chain contracts (double marginalization)

Variables: c supplier's production cost · w wholesale price · p, s, g retailer's price, salvage, goodwill · b buy-back price per return · r cost per return (shipping) · v supplier's salvage per returned unit (0 if scrapped) · π_R, π_S, π_{SC}.

Each firm decides with its own margin → retailer uses $C_u = p - w < p - c$, $C_o = w - s > c - s$ → orders **less** than the SC optimum. Contracts shift risk/margin to coordinate. Every profit uses the same chain: $Q \to z \to L(z) \to E[\text{short}] \to E[\text{sales}] \to E[\text{left}]$. **Order:** solve the retailer first (its Q, $E[\text{left}]$), then plug them into π_S.

Benchmark	Critical ratio and profits
Wholesale	$C_u = p - w + g$, $C_o = w - s_R \to Q_R$ $\pi_R = pE[\text{sales}] + s_R E[\text{left}] - gE[\text{short}] - wQ$ $\pi_S = (w-c)Q$
Integrated SC	$C_u = p - c + g$, $C_o = c - s_{\text{SC}}$ (w cancels) → larger Q $\pi_{\text{SC}} = pE[\text{sales}] + s_{\text{SC}}E[\text{left}] - gE[\text{short}] - cQ$

- Double-marginalization loss $= \pi_{\text{SC}} - (\pi_R + \pi_S)_{\text{wholesale}}$. Buy-back % $= b/w$.
- **Given b, find w^*** (column (a), $g = 0$): $w^* = p - (p-c)(p-b+r)/(p-s)$.

	(a) Returned · **retailer** pays return freight r · supplier salvages s_S	(b) Returned · **supplier** pays return freight r · supplier salvages s_S	(c) No return · retailer salvages s_R · supplier pays credit b
R^*	$R^* = p + g - \frac{(p-w+g)(p-s_{\text{SC}}+g)}{p-c+g}$, $s_{\text{SC}} = \max(s_R, s_S - r)$		
Retailer recovers R	$b - r$	b	$b + s_R$
b^* (given w: set $R = R^*$)	$R^* + r$	R^*	$R^* - s_R$
w^* (given b)	$p + g - \frac{(p-c+g)(p-b+r+g)}{p-s_{\text{SC}}+g}$	$p + g - \frac{(p-c+g)(p-b+g)}{p-s_{\text{SC}}+g}$	$p + g - \frac{(p-c+g)(p-b-s_R+g)}{p-s_{\text{SC}}+g}$
C_u^R	$p - w + g$		
$C_o^R = w - R$	$w - b + r$	$w - b$	$w - b - s_R$
α_R	$C_u^R/(C_u^R + C_o^R) = (p-w+g)/(p-R+g)$		
π_R	$pE[\text{sales}] + (b-r)E[\text{left}] - gE[\text{short}] - wQ$	$pE[\text{sales}] + bE[\text{left}] - gE[\text{short}] - wQ$	$pE[\text{sales}] + (b+s_R)E[\text{left}] - gE[\text{short}] - wQ$
π_S	$(w-c)Q - (b-s_S)E[\text{left}]$	$(w-c)Q - (b+r-s_S)E[\text{left}]$	$(w-c)Q - bE[\text{left}]$
Chain per leftover	$s_S - r$	$s_S - r$	s_R
Feasible b	$s_R + r < b < w$	$s_R < b < w$	$0 < b < w - s_R$

- $\pi_R + \pi_S = pE[\text{sales}] + (\text{SC per leftover})E[\text{left}] - cQ$. Equals integrated π_{SC} (with s) only if that value $= s$: in (a)/(b) need $v - r = s$, so with $v = s$ the SC is slightly below integrated (shipping r is wasted). (c) coordinates **exactly** (no shipping). Exact coordination for (a)/(b): use $s_{\text{SC}} = v - r$ in $(p-c)/(p-s_{\text{SC}})$.
- Who pays r does not change π_{SC}, only how it is split (b shifts by r).
- Accepted only if **both** π_R and π_S beat their wholesale-contract values.

Firm	Better off	No change	Worse off	Unsure
Retailer	$b - r > s_R$	$b - r \le s_R$	never	—
Supplier	$s_S \ge b$ (and $b - r > s_R$)	$b - r \le s_R$	—	$s_S < b$

Demand as a function of price: $D(p) = A - Bp$ (deterministic)

Variables: A demand at $p = 0$ · B units lost per €1 of price · $P_0 = A/B$ price at which demand hits 0 · c production cost · w wholesale · $m = p - w$ retailer margin · π^0 wholesale-contract profits · F franchise fee.

Each firm maximizes (its margin) × (demand). The **follower's** reaction is plugged into the **leader's** problem (solve backwards). Linear demand → optimal price is the **midpoint** between P_0 and the relevant cost.

Setting	Result
Retailer, given w	$\max_p (p-w)(A-Bp) \to p(w) = (P_0+w)/2$, $D = B(P_0-w)/2$
Supplier leads (sets w first)	$\max_w (w-c)B(P_0-w)/2 \to w = (P_0+c)/2$, $p = (3P_0+c)/4$, $D = B(P_0-c)/4$ $\pi_S = B(P_0-c)^2/8$, $\pi_R = B(P_0-c)^2/16$
Integrated	$\max_p (p-c)(A-Bp) \to p^I = (P_0+c)/2$, $D^I = B(P_0-c)/2 = 2D_{\text{dec}}$, $\pi^I = B(P_0-c)^2/4$
Retailer leads (sets $m = p - w$ first)	Supplier reacts: $\max_w (w-c)B(P_0-m-w) \to w(m) = (P_0-m+c)/2$, $D = B(P_0-m-c)/2$ Retailer: $\max_m m \cdot B(P_0-m-c)/2 \to m = (P_0-c)/2$ $\to w = (P_0+3c)/4$, $p = w+m = (3P_0+c)/4$, $D = B(P_0-c)/4$ $\pi_R = B(P_0-c)^2/8$, $\pi_S = B(P_0-c)^2/16$

- Decentralized: leader earns 2× the follower; total = 3/4 of integrated (25% lost). Price too high because both firms add a margin (double marginalization) → lower demand.
- **Quantity discount** (price w' only if $Q \ge D^I$): the retailer must charge p^I to sell D^I; both accept if $c + \pi_S^0/D^I \le w' \le p^I - \pi_R^0/D^I$.
- **Franchise** ($w = c$ + fixed fee F): retailer faces the chain's cost → sets p^I; both accept if $\pi_S^0 \le F \le \pi^I - \pi_R^0$.

8 · Order-up-to (periodic review, backorders)

Variables: L lead time (periods) · T review period ($T = 1$ default) · μ, σ demand per period · D_{L+1} demand over $L+1$ periods · S order-up-to (base-stock) level · α target in-stock · h holding/unit/period · b backorder penalty/unit · N periods · K cost per order · f freight cost per unit

Long-life product, stationary random demand, unlimited replenishments, backorders. Each period: order → the order from L periods ago arrives → demand. Order $= S - \text{IP}$ = last period's demand (one-for-one, pull). Level at end of period $= S - D_{L+1}$.

Each period (day, week, ...):

1. Order (order $= S - \text{IP}$, (= last period's demand)) → **2. Receive** (order placed L periods ago arrives) → **3. Demand** (random D; unmet → backorder)

Example, $L = 1$:

	Period 1	Period 2	Period 3
Order	q_1 placed	q_2 placed	q_3 placed
Receive	(from period 0)	q_1 arrives	q_2 arrives
Demand	D_1	D_2	D_3

An order placed in period t must cover demand in periods $t, \ldots, t+L$ → S protects $L+1$ periods.

Inventory level vs. inventory position

$$\text{IL} = \text{OnHand} - \text{BO} \qquad \text{IP} = \text{OnHand} + \text{OnOrder} - \text{BO} = \text{IL} + \text{OnOrder}$$

Event	OnHand	OnOrder	BO	IP
Place an order of q	—	$+q$	—	$\uparrow q$
Shipment of q arrives	$+q$	$-q$	—	unchanged
Arrival fills x backorders	$-x$	—	$-x$	unchanged
Demand d (stock / backorder)	—	—	$+$	$\downarrow d$

- **Never both positive** (arriving stock fills backorders first):
  - BO $> 0 \Rightarrow$ OnHand $= 0$
  - OnHand $> 0 \Rightarrow$ BO $= 0$

So:
- IL ≥ 0: OnHand $=$ IL, BO $= 0$
- IL < 0: BO $= -$IL, OnHand $= 0$
- IL and IP **can be negative**.
- $D_{t-1} = S - \text{IP}_t$
- $Q = S - \text{IP}$
- IP $= S$ right after ordering and right before demand
- **Only demand lowers IP; only ordering raises IP.** Arrivals and filling backorders just move units between terms.
- IP right after ordering $= S \to$ IP before ordering $= S -$ last period's demand.

	Review every period ($T = 1$)	**Review every T periods** (replace $L+1$ by $L+T$)
Policy		
Mean demand over horizon	$\mu_{L+1} = (L+1)\mu$	$\mu_{L+T} = (L+T)\mu$
Std. dev. (periods indep.)	$\sigma_{L+1} = \sqrt{L+1}\,\sigma$	$\sigma_{L+T} = \sqrt{L+T}\,\sigma$
Service level	$z = \Phi^{-1}(\alpha)$	
Cost-based α	$\alpha = \frac{C_u}{C_u + C_o} = \frac{b}{b+h}$ (**h per period**)	—
Order-up-to	$S = \mu_{L+1} + z\sigma_{L+1}$	$S = \mu_{L+T} + z\sigma_{L+T}$
Given S	$z = (S - \mu_{L+1})/\sigma_{L+1}$	$z = (S - \mu_{L+T})/\sigma_{L+T}$
Service		
In-stock prob.	$P(D_{L+1} \le S) = \Phi(z)$	$P(D_{L+T} \le S) = \Phi(z)$
Stockout prob.	$1 - \Phi(z)$	
$E[\text{Backorder}]$	$\sigma_{L+1}L(z)$	$\sigma_{L+T}L(z)$
Fill rate	$1 - E[\text{BO}]/\mu$	$1 - E[\text{BO}]/(\mu T)$
Target fill rate FR → $L(z)$	$L(z) = (1-\text{FR})\mu/\sigma_{L+1}$	$L(z) = (1-\text{FR})\mu T/\sigma_{L+T}$
Stockouts in N (review cycles)	$N(1-\alpha)$	
Cycles until 1st stockout	$1/(1-\alpha)$	
Inventory		
$E[\text{On-hand}]$ = SS	$S - \mu_{L+1} + E[\text{BO}]$ $= \sigma_{L+1}(\varphi(z) + z\Phi(z))$	$S - \mu_{L+T} + E[\text{BO}]$ $= \sigma_{L+T}(\varphi(z) + z\Phi(z)) \approx z\sigma_{L+T}$
Cycle inventory	$\mu/2$	$\mu T/2$
Avg. inventory	SS $+ \mu/2$	SS $+ \mu T/2$
$E[\text{On-order}]$ = pipeline	$L\mu$ (Little's law; L, not $L+T$; indep. of σ)	
Days of supply	SS $/\mu$ (+ cycle 1/2)	SS $/\mu$ (+ cycle $T/2$)
Cost (keep all in the same time unit)		
Orders / yr	$n = 1/T_{\text{yr}}$	$n = (1 - P(D_T = 0))/T_{\text{yr}}$ ($= 1/T_{\text{yr}}$ for normal demand)
Annual cost	$\underbrace{(\text{SS} + \text{CI})H_{\text{yr}}}_{\text{holding}} + \underbrace{nK}_{\text{ordering}} + \underbrace{L_{\text{yr}}\mu_{\text{yr}}H}_{\text{if firm owns in-transit}} + \underbrace{f\mu_{\text{yr}}}_{\text{transport}}$	

- Rounding: always recompute z from the **rounded** S.
- $E[\text{BO}] \approx$ newsvendor expected lost sales, but the fill rate divides by **one review cycle's** demand (μT), not μ_{L+T}.
- Optimal in-stock is usually high because $b \gg h$. Annual $h \to h/52$ for weekly periods.
- Safety-stock drivers: lead time, variability σ/μ, service level. $L \downarrow \Rightarrow \mu_{L+T}, \sigma_{L+T}, S$, on-order and SS all $\downarrow$ (strongest at high α).
- Target fill rate: $L(z)$ from the table → look up z (larger) → S.
- **Choosing T** (min ordering + holding per yr):
  1. For each $T = 1, 2, 3, \ldots$: μ_{L+T}, $\sigma_{L+T} \to S \to E[\text{BO}] = \sigma_{L+T}L(z) \to \text{SS} = S - \mu_{L+T} + E[\text{BO}]$.
  2. $\text{Cost}(T) = H_{\text{yr}}(\text{SS} + \mu T/2) + Kn$, $n \approx 1/T_{\text{yr}}$ (+ $L\mu H$ if pipeline owned; transport $f\mu_{\text{yr}}$ is the same for every T).
  3. Pick the min. Start near $T_{\text{EOQ}} = \sqrt{2K/(H\mu)}$ (in periods); the optimum is close but not equal (EOQ ignores SS).
  - $T \uparrow$: fewer orders, but SS $\propto \sqrt{L+T}$ and cycle stock $\propto T \uparrow$. If $\alpha = b/(b+h)$ with h per review period, a longer T gives a lower α.